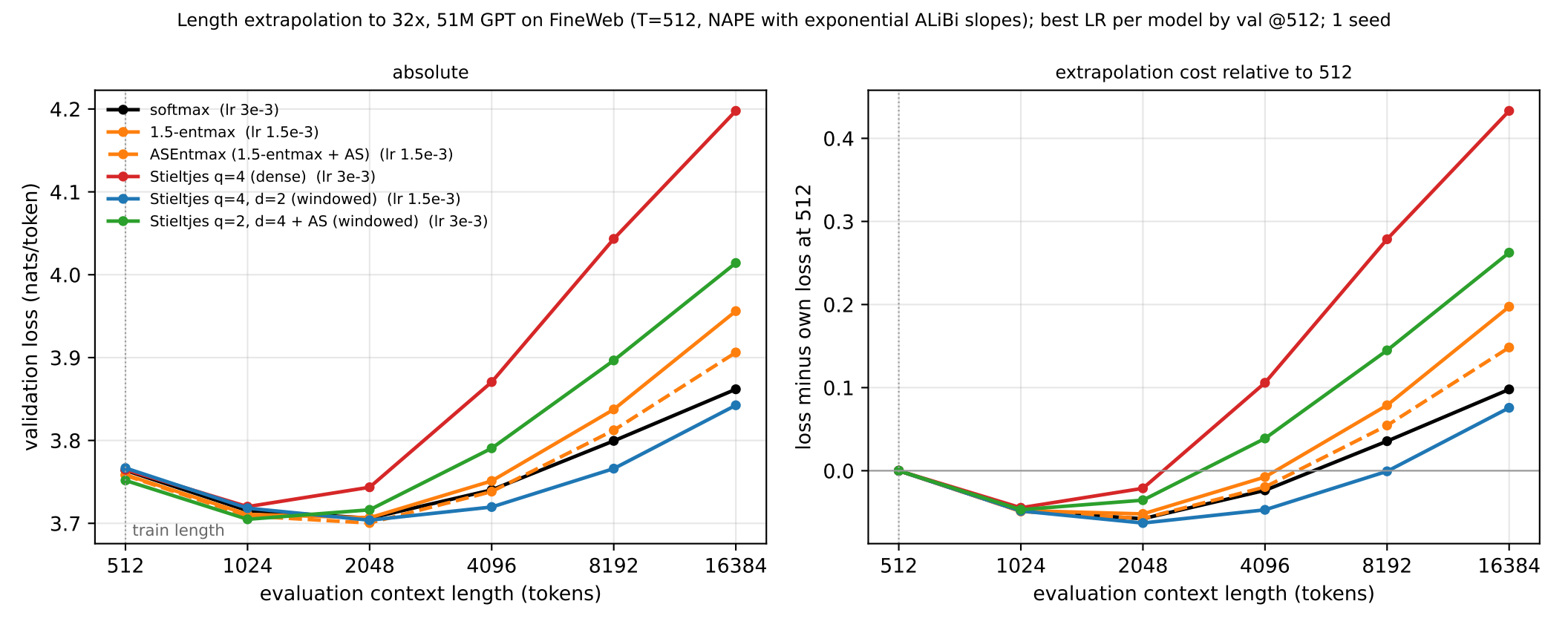
Length extrapolation to 32x, 51M GPT on FineWeb (T=512, NAPE with exponential ALiBi slopes); best LR per model by val @512; 1 seed
absolute
extrapolation cost relative to 512
validation loss (nats/token)
loss minus own loss at 512
evaluation context length (tokens)
evaluation context length (tokens)
softmax (lr 3e-3)
1.5-entmax (lr 1.5e-3)
ASEntmax (1.5-entmax + AS) (lr 1.5e-3)
Stieltjes q=4 (dense) (lr 3e-3)
Stieltjes q=4, d=2 (windowed) (lr 1.5e-3)
Stieltjes q=2, d=4 + AS (windowed) (lr 3e-3)
train length
4.2
4.1
4.0
3.9
3.8
3.7
0.4
0.3
0.2
0.1
0.0
512
1024
2048
4096
8192
16384